Exercise	Topic, Focus & SLOs	Portfolio activities
P1	**SLO4** **AC4.1** **Input & Output /** **math Module**	**P1 Stadium Screens** **Your Task:** A sports stadium has made two spaces available and hopes to install two new rectangular video screens. The stadium has: • Space for one **large screen**, with a maximum diagonal length of **12 metres**. • Space for one **small screen**, with a maximum diagonal length of **7.52 metres**. The larger suitable screen should always be allocated to the large-screen space. A second screen can only be installed if its diagonal does not exceed 7.52 metres. Write a Python program that allows stadium staff to enter the dimensions of two proposed screens and determine whether each screen is suitable for installation. Your program must: • Input the height and width of both proposed screens. • Calculate the diagonal length of each screen using Pythagoras' theorem. • Determine whether each screen fits in one of the available spaces. • Display how many screens will fit and clearly identify which screen or screens are suitable. • Display clear and helpful messages for the stadium staff. Your program must also: • Use clear and meaningful variable names. • Include comments explaining each main part of the program.

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- Follow good formatting practices, including consistent indentation and spacing.

A diagonal can be calculated using:

$$\text{diagonal} = \sqrt{\text{height}^2 + \text{width}^2}$$

Test Data

You should use the data below to test your solution.
You must carry out at least the tests shown below and provide evidence of the results to show your solution works as expected.

Test	Screen 1 dimensions	Screen 2 dimensions	Expected result
1	Height: 15 m, Width: 8 m	Height: 10.5 m, Width: 5.6 m	One screen fits: Screen 2 fits in the large-screen space
2	Height: 10 m, Width: 6 m	Height: 7 m, Width: 2.5 m	Both screens fit
3	Height: 13 m, Width: 5 m	Height: 8 m, Width: 4 m	Neither screen fits

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